CSE 4711: Artificial Intelligence - Quiz #4 **20 November 2023**
Duration: 25 Minutes **Full Marks: 15**
Student ID:

Sometimes MDPs are formulated with a reward function $R(s,a)$ that depends on the action taken or with a reward function $R(s,a,s')$ that also depends on the outcome state.

1. Modify the Bellman Equation to determine the value of sate based on these formulations. 4 (CO2) (PO2)

Solution:

For $R(s,a)$, we have:

$$V(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} T(s,a,s')V(s') \right]$$

And for $R(s,a,s')$, we have:

$$V(s) = \max_a \sum_{s'} T(s,a,s') \left[R(s,a,s') + \gamma V(s') \right]$$

2. Show how an MDP with reward function $R(s,a,s')$ can be transformed into a different MDP with reward function $R(s,a)$, such that optimal policies in the new MDP corresponding exactly to optimal policies in the new MDP correspond exactly to optimal policies in the original MDP. 5 (CO1) (PO1)

Solution:

There are a variety of solutions here. One is to create a "pre-state" $pre(s,a,s')$ for every s, a, s', such that executing a in s leads not to s' but to $pre(s,a,s')$. In this state is encoded the fact that the agent came from s and did a to get here. From the pre-state, there is just one action b that always leads to s'. Let the new MDP have transition T', reward R', and discount γ'. Then

$$\begin{aligned}
&T'(s,a,pre(s,a,s')) = T(s,a,s') \\
&T'(pre(s,a,s'),b,s') = 1 \\
&R'(s,a) = 0 \\
&R'(pre(s,a,s'),b) = \gamma^{-\frac{1}{2}} R(s,a,s') \\
&\gamma' = \gamma^{\frac{1}{2}}
\end{aligned}$$

3. Now do the same to convert MDPs with $R(s,a)$ into MDPs with $R(s)$. 6 (CO1) (PO1)

Solution:

In keeping with the idea of previous answer, we can create states $post(s,a)$ for every s, a, such that

$$\begin{aligned}
&T'(s,a,post(s,a,s')) = 1 \\
&T'(post(s,a,s'),b,s') = T(s,a,s') \\
&R'(s) = 0 \\
&R'(post(s,a,s')) = \gamma^{-\frac{1}{2}} R(s,a) \\
&\gamma' = \gamma^{\frac{1}{2}}
\end{aligned}$$